Internship offer at LIMOS
Clustering on spectral imaging for tumor detection

Context and objective:

Monitoring the dynamics of cancer tissues using non-invasive techniques represents crucial complement to study and predict the efficacy of therapeutic intervention. Herein, the immune system, e.g. tissue infiltrating immune cells and immune cell activation, play an important role for treatment susceptibility. Among non-invasive methods, observing the transition of cell state using Raman imaging techniques has shown promising results.

Over the past few years, machine learning has gained considerable traction for the analysis of Raman data [1,3,4]. Nonetheless, despite significant advances, most methods remain focused on supervised tasks where a model is learned assuming knowledge of target species and their values [6]. In particular, approaches for compositional characterization of Raman measurements typically use explicit information provided during model training about the presence (or absence) of particular target species [9] or known concentrations [2,5,8]. In this context, it is clear that the development of methods that do not rely on known target substances and concentrations remains essential.

Raman imaging techniques produce images where each pixel represents an individual Raman spectrum, recapitulating a cell's or tissue's molecular composition at spatial resolution. The Raman spectrum shows the Raman shift varying with the intensity. Although it is not a time series, it represents a sequence with a dedicated order. The structure of the Raman data, as well as the mix of biological components in the Raman signature, make the data particularly difficult and elaborate to analyze, currently limiting its clinical translation.

The objective of this internship is to analyze Raman-based biomarkers in order to discriminate between groups of cells, such as macrophage phenotype and dendritic cell. The steps of the internship include:

1. The exploration / visualization of the data to reveal the predominant characteristics of the dataset (noise, missing values, size).
2. The use of a clustering algorithm that handles ordered series and multi-class labels, by implementing evidential clustering conjugated with a Dynamic Time Warping dissimilarity [7].

Required skills: machine learning, data mining, python

Duration : 6 months, starting anytime between January 1, 2027 and March 1, 2027

Location: The internship will take place at LIMOS, Aubière, France. Regular visioconférences will occur with the supervisors located in Saint Etienne and Tübingen.

Salary: regulatory compensation

Contacts:

- Violaine Antoine : violaine.antoine@uca.fr (LIMOS)
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Application: Applications must be submitted by email to Violaine Antoine and should include a detailed CV, as well as the grades obtained during the most recent academic year.

Bibliography:

[1] X Bi, L Lin, Z Chen, J Ye. Artificial intelligence for surface-enhanced Raman spectroscopy. Small Methods 8 (2024).
[2] OC Koyun et al. RamanFormer : a transformer-based quantification approach for Raman mixture components. ACS Omega 9 : 23241-23251 (2024).
[3] R Luo, J Popp, T Blocklitz. Deep learning for Raman spectroscopy: a review. Analytica 3: 287-301 (2022).
[4] F Lussier, V Thibault, B Charron, GQ Wallace, JF Masson. Deep learning and artificial intelligence methods for Raman and surface-enhanced Raman scattering. Trends Anal. Chem. 124 (2020).
[5] L Pan et al. Identification of complex mixtures for Raman spectroscopy using a novel scheme based on a new multi-label deep neural network. IEEE Sens. J. 21: 10834-10843 (2021).
[6] L Pan, P Zhang, C Daengngam, S Peng, M Chongcheawchamnan. A review of artificial intelligence methods combined with Raman spectroscopy to identify the composition of substances. J. Raman Spectrosc. 53: 6-19 (2022).
[7] A. Soubeiga, V. Antoine, S. Moreno, J. Koko. Evidential clustering with view-weight learning for proximity data. Neurocomputing, 131886, 2025.
[8] Z Wang et al. Deep learning-based Raman spectroscopy qualitative analysis algorithm: a convolutional neural network and transformer approach. Talanta 275 (2024).
[9] Z Zhao et al. ConInceDeep : a novel deep learning method for component identification of mixture based on Raman spectroscopy. Chemom. Intell. Lab. Syst. 234 (2023).